

Regional resilience assessment based on city network risk propagation and cooperative recovery

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ABSTRACT

With the rapid development of transportation systems and increased connectivity between cities, the capacity of regional cities to propagate risks and cooperate with each other has a great impact on regional resilience. Given that a region can be seen as a networked and multidimensional system with complex interactions of cities, it is necessary to assess regional resilience based on the complex network thinking. This paper views the region as a two-layer coupled city network with risk propagation and cooperative recovery and propose the regional resilience assessment framework. Then, a network of 337 prefecture-level cities in mainland China is used as the study area to assess the impact of different cities on the regional resilience after a sudden public health disaster. The results show that cities in western China have less impact on regional resilience, while cities in central and eastern China have more impact. In addition, we found that each indicator under the three dimensions of regional network resilience has different degrees of correlation with influencing factors. Meanwhile, the changes in the same influencing factor may cause indicators under different dimensions of resilience to increase or decrease. There is a trade-off between approaches to improve regional resilience under each dimension.

1. Introduction

In the context of global climate change, large-scale disasters are more frequent and regional development is facing great threats (Monstadt & Schmidt, 2019). Regional resilience, which describes the ability of regions to respond and recover in a timely manner in the face of disasters, has been emphasized and focused on by governments and the academic and industrial sectors (Cutter et al., 2010). Meanwhile, the rapid development of modern transportation systems has led to closer links between cities in the regions and the formation of the network-scale (Wei & Xiu, 2020). The gradual formation of city networks greatly facilitates the movement of people, materials, and commerce within the region. When the region suffers a disaster attack, the risk propagation and cooperative recovery based on the city network will make the performance of regional material operation and infrastructure security show a dynamic process, which has a significant impact on regional resilience.

The connectivity of regional city networks further facilitates the flow of resources from one subsystem to another, but risks also spread and have a magnifying effect (Li et al., 2021), which means that natural or

man-made disasters are likely to have a greater impact on the normal functioning of the region and human activities (Wang et al., 2019). For example, in 2021, the “July 20” rainstorm resulted in the shutdown of the Zhengzhou Railway, which in turn led to massive disruptions and delays in railway traffic throughout China, causing significant losses nationwide (Wu et al., 2022). In the case of 2019 Manhattan blackout, an emergent power outage hit the West Side of Manhattan resulted in the disruption of power system networks throughout the 30-block area from Times Square to 72nd Street and from 5th Avenue to the Hudson River, leaving about 73,000 customers in the region without electricity. This power outage disaster similarly exemplifies the expanding disaster impact of the propagation of disaster risk in the city networks (Li, Ma, & Cao, 2020; Li, Zobel, et al., 2020).

However, the connectivity of city networks also provides new ideas for disaster response, such as cooperative recovery (Kapucu & Garayev, 2013; Kapucu & Garayev, 2014). Regions can carry out cooperative emergency response relying on the city network and integrate the resources of various cities in the region to rescue damaged areas, which can effectively deal with attacks and complete recovery. For example, in

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the face of the COVID-19 epidemic in 2020, China quickly integrated medical resources from all cities in the country to support Hubei Province, alleviating the collapse of the medical system in Hubei province and effectively reducing the losses caused by the epidemic (Shan et al., 2023). Similarly, when Philippines suffered the impact of Typhoon Haiyan, the cooperative relief network based on what was constructed by countries around the globe covered the local government's shortcomings, addressed the shortage of materials and effectively reduce the disaster losses (Santiago et al., 2016).

As the size of the regional city network continues to grow, the trade-off impact of risk propagation and cooperative recovery will make the regional resilience more complex, especially for the dynamic change of regional resilience under the impact of sudden public health disasters. Sudden public health disasters are characterized by the suddenness of outbreaks, the widespread nature of transmission, the uncertainty of impacts, and the comprehensive nature of governance (Afolabi, 2018). In recent years, the COVID-19, Ebola, and SARS epidemics all have spread rapidly in various cities among the region, causing significant impacts on the whole region and even globally. For the emergency response to the sudden public health disaster, only relying on the efforts of outbreak city cannot fulfill the supply of medical resources and the blockage of epidemic risk after the disaster. Regional government administrators usually use the flow of people, money, and materials between city networks to carry out coordinated recovery and to jointly resist the impact of sudden public health emergencies and ensure regional resilience. Assessing the regional resilience capacity under the dynamic process of responding to sudden public health disaster attack is of great practical significance for regional disaster reduction and sustainable development.

Therefore, the significance of this study is as follows: (1) To model the city network under risk propagation and cooperative recovery, (2) To quantify the state of change of regional resilience capacity under the sudden public health disaster attack, (3) To cluster the cities among the network for further sudden public health disaster reduction, (4) To identify the impact of different elements of the regional city network on its resilience.

The rest of the paper is organized as follows: Section 2 reviews related literatures. Section 3 proposes the regional city network resilience assessment method. In Sections 4 and 5, we select mainland China as the research object and conduct sudden public health disaster simulation to assess its resilience with the impact of risk propagation and cooperative recovery based on the city network. Section 6 is the summary of the whole article.

2. Literature review

2.1. Regional resilience

The term resilience originated from the Latin word “resilio” and was first used to measure ecological resilience in the 1950s (Arani et al., 2021; Folke et al., 2004). The concept of resilience has been extended to different contexts, including engineering resilience (David, 2007; Wildavsky, 1988), economic resilience (Raco & Street, 2012; Rose & Lim, 2002), ecological resilience (Holling, 1973; Ludwig, 1996) and social resilience (Adger, 2000; Saja et al., 2018), and its connotation has been further deepened.

The increase or perceived increase in the number of shocks and disruptions such as natural disasters and sudden public health disasters led to the rising interest in resilience from the regional perspective (Gong et al., 2020). Unlike traditional regional security studies that focus on reducing interference, the resilience perspective puts more emphasis on the ability of a region to absorb blows and complete recovery under the impact of risks (Peng et al., 2015). The regional resilience definition emphasizes the ability of a region to anticipate, prepare for, respond to and recover from a disturbance (Bristow, 2010; Foster, 2007).

Aside from conceptual identification (Sharma et al., 2023), the

studies on regional resilience has led to many empirical papers, including construction of better resilient regions (Pickett et al., 2014), improvements in resilience assessment methods (Jabareen, 2013), and investigation of factors influencing resilience (Chen et al., 2021). However, most of the studies have been conducted by considering only a single small region or a single city (Wang, Xu, & Wang, 2023a; Wang, Xu, & Wang, 2023b; Wang, Zhao, & Wu, 2023). In reality, the expansion and acceleration of various infrastructural constructions have led to increasing interaction effects among different cities in the whole region (Xu et al., 2023), which have a significant impact on overall regional resilience. Extending the concept of resilience to the whole regional scale and considering interactions between cities are necessary to improve the overall capacity of region systems to deal with disaster crises (Wang et al., 2018).

2.2. City network resilience

Globalization and informationization have dramatically changed the spatial organization of the region in today's world, the original vertical geospatial structure is replaced by the form of networks (Derudder & Neal, 2018). The development of flow space theory and social network analysis methodologies has substantially expanded the study of urban and regional networks. (Pan et al., 2019). This calls for a broadening of the scope of research on regional resilience and further consideration of the impact on regional resilience of city networks formed by the interactions between cities within a region. City network resilience began to get attention.

In terms of network construction methods, research on regional network resilience consider cities as nodes and edges as the relations between different cities, constructing different types of regional networks with entities or abstract data streams. As shown in Table 1, researchers have explored the resilience capacity of the region city network in a variety of aspects, including economic, ecological, infrastructure and social (Wei & Xiu, 2020), covering regional scales ranging from a cluster of cities to an entire country or even a global region. For example, Ye and Qian (2021) introduced a modified gravity model to construct the city economic network in the region of Guanzhong Plain City Cluster, exploring the key cities in the region that influence economic resilience, as well as the development of resilience capacity in the whole region. Shughrue and Seto (2018) developed a global synoptic network of industrial flows among 16 sectors and all 1686 urban areas with >300,000 inhabitants to explore the impacts of climate change on the resilience of urban areas globally. However, most of the above studies have only considered a single aspect of regional resilience, ignoring the impact of multiple elements among city networks on comprehensive regional resilience. With the deepening of regional resilience research, there is a greater emphasis on multidimensionality and systematicity, stressing the incorporation and interaction of multidimensions among cities (Sharifi, 2019; Wang et al., 2018). This requires considering the city network as a multidimensional network and synthesizing regional resilience research under the influence of multiple aspects. (See Table 3.)

2.3. Quantitative measurement of resilience

For the quantitative measurement of the resilience of regional city networks, complex network approach has been applied to the study of regional resilience (Crucitti et al., 2006; Liu et al., 2022). Researchers have mostly portrayed the resilience of regional networks from two perspectives: static and dynamic (Dabson et al., 2012; Wilbanks, 2008).

The static perspective considers regional network resilience capacity as a fixed value, and its quantitative measurement is mostly through the method of indicator systems. For example, The Rockefeller Foundation has proposed a system of resilience indicators comprising four dimensions including health and well-being, economy and society, infrastructure and environment, and leadership and strategy to assess 100

Table 1

Framework for resilience assessment of regional city networks.

Aspects	Study area of network	Reference	Network construction	
			physical	abstract
Economic	City relation network in Harbin-Dalian urban belt	(Xie et al., 2020)		✓
	Urban economic linkage network in Guanzhong Plain	(Ye & Qian, 2021)		✓
	City Cluster Supply chain network in Taian	(Haraguchi & Lall, 2015)	✓	
Ecological	Chinese provinces city network	(Wang, Xu, & Wang, 2023a; Wang, Xu, & Wang, 2023b; Wang, Zhao, & Wu, 2023)		✓
	Global CO2 transfer network	(Zhang et al., 2017)	✓	
	Ecological Function Network in Taiwan Island	(Chang et al., 2007)		✓
Infrastructure	Global urban-industrial network	(Shughrue & Seto, 2018)		✓
	Infrastructure network of Brazilian city of Salvador's	(Silva et al., 2022)		✓
	Electricity transmission network in Netherlands	(Bollinger & Dijkema, 2016)	✓	
Social	Bonding and bridging networks on the Bangladeshi coast	(Islam & Walkerden, 2014)		✓
	Ecosystem network of the city of Sarno, Italy	(Bozza et al., 2017)		✓

Table 3

Network resilience indicator.

Dimensions	Resilience indicator	Generalized formula
Robustness loss	ΔQ	$\Delta Q = 1 - Q_{min}$
Recovery time	t^*	$Q(t^*) - Q(t^* - 1) = 0$
Loss of quality	L	$L = \int_0^T [1 - Q(t)] dt$

resilient cities in the global network. (Spaans & Waterhout, 2017). Meanwhile, some network structure metrics such as degree, betweenness, clustering, etc. also have been widely used to construction of the indicator system to measure network resilience (Dumedah & Garsonu, 2021; Li et al., 2017; Meng et al., 2020; van Nes & Bovy, 2000). However, when the region faces with external disruptions, the factors flow of people, logistics, capital, technology information and even risk will show the obvious characteristics of mutual interference or collaboration and complementarity among city network. Simply portraying a single resilience capacity value through an indicator system does not show the dynamic impact of these flowing elements on resilience under external disruptions. As a complex system of multiple interacting elements, its factor connection (Mkinen, 2001) and space-time coordination (Meijers, 2005) also have a huge impact on resilience, and further attention needs to be paid to the dynamics of network resilience after a disaster strike.

In the existing research on dynamic perspectives, the region network resilience is seen as the variables of the dynamic process (Simmie & Martin, 2010). Jiang et al. (2022) introduced Bayesian Dynamic Network Analysis to divide the dynamic change of the region network

into five stages, including preparation, resistance, adaptation, recovery and evolution and describes the network resilience in each phase in turn. Zhang et al. (2023) use the change of vehicle travel speed to portray the dynamic change of resilience of the road network at each specific time. For the resilience of regional city networks, quantitatively exploring the resilience at each moment under such strikes is more in line with the needs of realistic scenarios. However, researches on resilience in the perspective of regional networks under strikes is still in the infancy, and the changes in the elements of resilience impact considered are also only one-sided.

2.4. Risk propagation and cooperative recovery

In the face of the disaster attack, the impact of city network systems on regional resilience is two-fold. Either its interaction and dissemination facilitate the propagation of crises and generates risk propagation in the region, or its pattern of collaboration and complementarity resists external shocks and safeguards regional resilience. The trade-off impact between risk propagation and cooperative recovery has been found in the study of supply chain network resilience (Li, Ma, & Cao, 2020; Li, Zobel, et al., 2020).

When the orderly flow of various production factors among cities in city networks greatly enhances regional production efficiency, risk factors are also transmitted and spread along the network. For example, in April 2016, the strong earthquake occurred in Kumamoto Prefecture, Japan, leading some local auto parts producers to suffer a shutdown, and the resulting supply chain disruption caused a significant nationwide reduction in auto production (Wagner & Silveira-Camargos, 2012). Some researchers argue that some risks with strong propagation effects (e.g., cascading failures, etc.) which are themselves born out of the highly connected global network (Helbing, 2013). Others believe that when a region faces crises and disturbances, it is highly likely to induce other related crises, and the disruption of local functions may lead to the non-functioning of other parts. So, a crisis occurring in one region is likely to generate risk propagation in a closely connected city network, triggering more serious secondary crises (Beck & Villarroel, 2013). The appearance of risk propagation will cause dynamical changes in the network tend to be fast, potentially outstripping the rate at which one can learn about the characteristic system behavior, or at which humans can react (Helbing, 2013). Likewise, risk propagation means that one event can trigger further events, thereby creating amplification and cascade effects, which implies a large vulnerability to perturbations, variations or random failures (Buldyrev et al., 2010; Gao et al., 2011; Watts, 2002).

Meanwhile, the region is not passively disturbed after suffering disaster attacks, but also shows some initiative (Peng et al., 2015). The cities in the region can rely on the city network to integrate the overall resources for cooperative recovery to external disturbances. For example, the effective containment of the novel coronavirus pneumonia epidemic in China in 2020 was a result of the establishment of an inter-city public health network and a network of health care workers (Wang, Xu, & Wang, 2023a; Wang, Xu, & Wang, 2023b; Wang, Zhao, & Wu, 2023). In the 2005 Hurricane Katrina disaster in the United States, the dysfunctional relationship between state and local governments and the failure to establish effective coordinative networks among cities led to catastrophic consequences (Abbott, 2007). If the cities in the region can cooperate in emergency response, they can act like resilient and tense spider webs to absorb disruptions and shocks, ensuring regional environmental safety and order.

These studies show that researchers have recognized the existence of the risk propagation and cooperative recovery action in city networks under the disaster attack. However, the quantitative analysis of the impact of this mechanism of action on regional resilience is still at a blank stage.

Through the above literature, regional resilience has always been a key issue for regional development. However, most studies have only considered a single small region or a single city, ignoring the impact of

the city network formed by the interactions between cities on regional resilience. In the perspective of resilience research on regional city networks, most of the existing studies only consider the resilience of a single element between cities, such as economy, transport, etc. However, after the real disaster attack like sudden public health disasters, there is a coupling of risk propagation and cooperative recovery based on city networks, which will have a trade-off effect on the resilience assessment. In addition, for the measurement of regional resilience, the static resilience analysis through the method of indicator system is still the mainstream. But, for the regional resilience after disaster attacks is a dynamic process, and it is necessary to further improve the dynamic quantitative analysis of the resilience capacity. In this paper, we consider the overall regional resilience capacity in the context of inter-city interactions, and portray the impact of risk propagation and cooperative recovery in the city network with a two-layer coupled network model, as well as plot the resilience curve to quantitative assess the dynamic changes of regional resilience with robustness loss, recovery time and loss of quality.

3. Method

In this section, we construct a resilience assessment methodology framework as shown in Fig. 1, including three parts: risk propagation and cooperative recovery coupling network construction, city state transition under the sudden public health disaster attacks, and the metrics of network resilience.

3.1. Risk propagation and cooperative recovery coupling network construction

The whole region can be regarded as a two-layer coupled city network. Due to the movement of people in the city network after the sudden public health disaster attack, there will be the risk propagation mechanism, existing a risk propagation network $G^a(V, E^a)$; while as cities in the region work together on emergency response, there will also be the cooperative recovery mechanism, existing a cooperative recovery network $G^b(V, E^b)$. $V = \{v_1, v_2, v_3, \dots, v_n\}$ is the set of nodes in the network, representing the cities in the region. E^a and E^b are the sets of connected edges in the two-layer coupled network, where edge $e_{ij}^a \in E^a$ denotes the risk propagation action between node v_i and node v_j , and edge $e_{ij}^b \in E^b$ denotes the cooperative recovery action between node v_i and node v_j .

Cities damaged by disasters are more probably to transmit risk to neighboring cities along their spatial boundaries. Risk propagation networks $G^a(V, E^a)$ are then constructed based on the spatial relationships of cities. While a city's recovery process from disaster damage can be assisted by all other cities. Thus, the cooperative recovery network $G^b(V, E^b)$ is fully connected. For example, there exists a two-layer coupled network composed of 9 nodes in a certain region, and its risk propagation and cooperative recovery mechanism of action is shown in Fig. 2. In the risk propagation network $G^a(V, E^a)$, node 6 has spatial adjacency with 3 nodes (nodes 3, 4, 5), so it may be affected by risk transmission from $e_{36}^a, e_{46}^a, e_{56}^a$; while in the cooperative recovery network $G^b(V, E^b)$, node 6 can receive medical assistance from all other nodes in the network through coordinative emergency response, and there are a total of 8 edges with cooperative recovery function.

3.2. City state transition under disaster

In the face of the sudden public health disaster attack, each individual city in the region is usually regarded as the direct disaster bearing body, which usually has three states: normal (vulnerable), normal (immune) and damaged. At a certain period t , the cities in normal state ($x_i(t) = 0$) may be vulnerable cities, or they may be immune and have

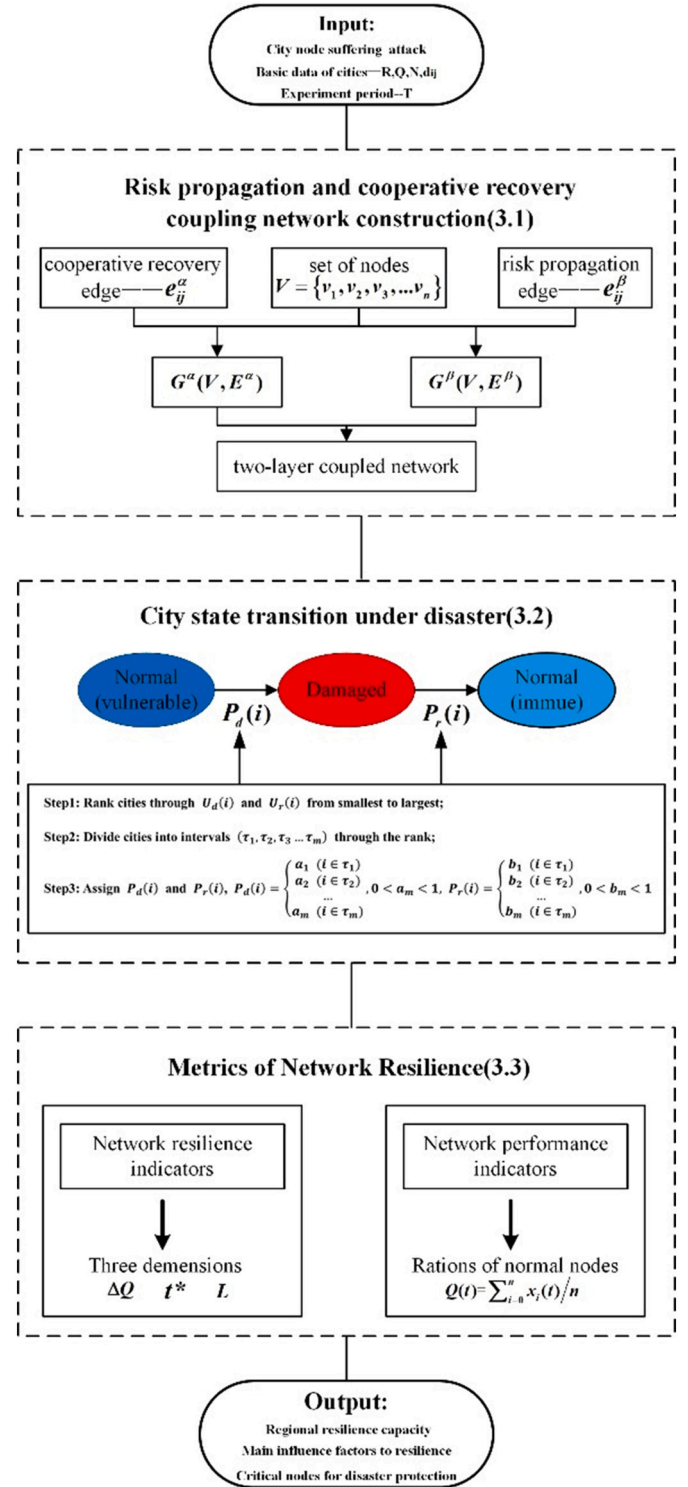


Fig. 1. Framework for resilience assessment of regional city networks

resistance to the sudden public health disaster attack such as infectious disease. Fig. 3 shows the transition between the three states among cities in the city network. Any city i ($x_i(t) = 0$) in the normal (vulnerable) state may turn into a damaged city ($x_i(t+k) = 1$) at some point later influenced by risk propagation. The probability of transition from normal (vulnerable) to damaged state is called damaged probability $P_d(i)$. At period $t+k$, the damaged city i ($x_i(t+k) = 1$) may also rely on cooperative recovery within the region to return to the normal(immune) state ($x_i(t+m) = 0, m > k$), and this probability of transition from

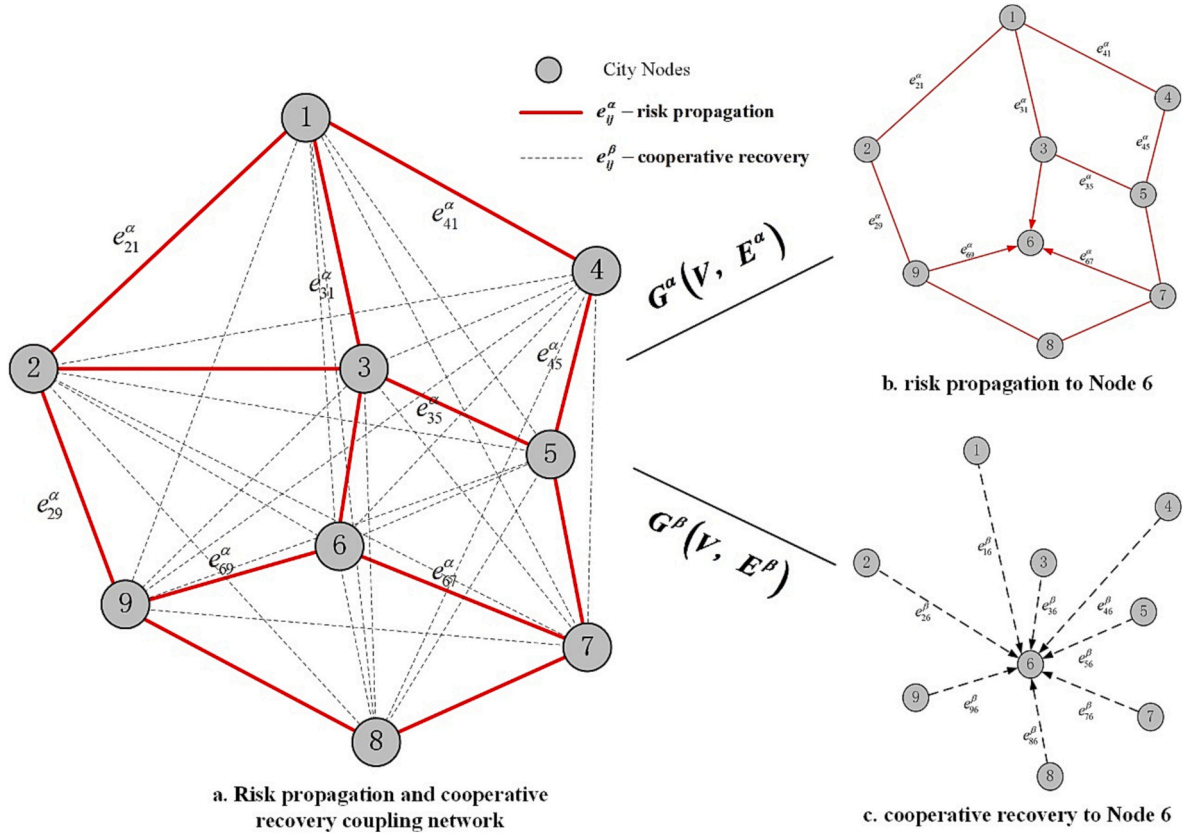


Fig. 2. Mechanisms of action for risk propagation and cooperative recovery

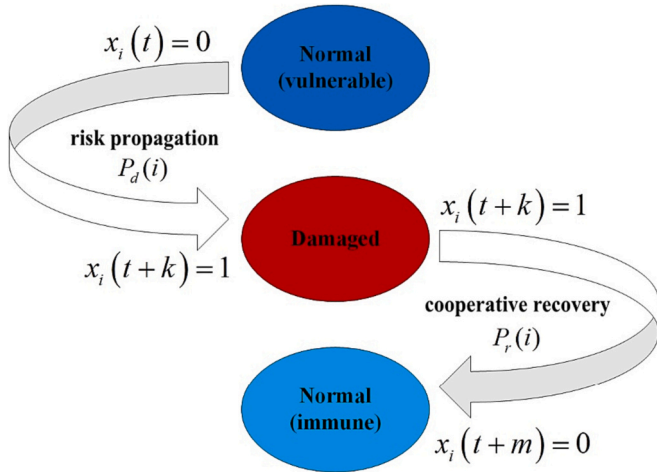


Fig. 3. City state transition.

damaged state to normal (immune) state is called recovery probability $P_r(i)$. As soon as the city i transition from its damaged state to normal (immune), it will gain resistance, after which $P_d(i) = 0$.

Due to risk propagation and cooperative recovery mechanism in the regional city network, the damage probability $P_d(i)$ and recovery probability $P_r(i)$ of each city node v_i will be comprehensively affected by the e_{ij}^α and e_{ij}^β edges. Their quantitative calculation process includes the following three steps:

- Step1: Rank cities through $U_d(i)$ and $U_r(i)$ from smallest to largest;
 Step2: Divide cities into intervals $(\tau_1, \tau_2, \tau_3 \dots \tau_m)$ through the rank;

Step3: Assign $P_d(i)$ and $P_r(i)$, $P_d(i) = \begin{cases} a_1 & (i \in \tau_1) \\ a_2 & (i \in \tau_2) \\ \dots & \\ a_m & (i \in \tau_m) \end{cases}$, $0 < a_m < 1$,

$$P_r(i) = \begin{cases} b_1 & (i \in \tau_1) \\ b_2 & (i \in \tau_2) \\ \dots & \\ b_m & (i \in \tau_m) \end{cases}, 0 < b_m < 1;$$

$U_d(i)$ and $U_r(i)$ respectively represent the potential impact of risk propagation and cooperative recovery on node v_i . As shown in Eqs. (1) and (2), the impacts depend on the summation of the propagation intensity w_{ij}^α of each edge e_{ij}^α and the cooperative recovery intensity w_{ij}^β of each edge e_{ij}^β in the network.

$$U_d(i) = \sum_{j=1}^n w_{ij}^\alpha x_j \quad (1)$$

$$U_r(i) = \sum_{j=1}^n w_{ij}^\beta \quad (2)$$

Gravity model is used to describe the impact between spatial objects (Whitehead et al., 1995). This paper selected the distance between cities d_{ij} , the number of urban population R , the supply of urban medical resources S and the demand N as related parameters, and quantitatively calculated urban propagation intensity w_{ij}^α and recovery intensity w_{ij}^β based on the gravity model, as shown in Eqs. (3) and (4):

$$w_{ij}^\alpha = \frac{R_i R_j}{d_{ij}^2} \quad (3)$$

$$w_{ij}^\beta = \frac{W_i S_j}{d_{ij}^2} \left(W_i = \frac{1}{N_i} \right) \quad (4)$$

2.1 Metrics of network resilience

As shown in Fig. 4., the city network performance $Q(t)$ will fluctuates during the 0-T period after the disaster attack. When $t = 0$, a city node of the network suffers the disaster attack, and the whole network performance begins to decline under risk propagation. When $t = t_{min}$, the network goes to the most severe state and the its performance drops to the lowest point Q_{min} . While with the action of the cooperative recovery mechanism in the network, the network performance gradually recovers. When $t = t^*$ (which t^* indicates a moment of recovery stability before T), the network performance $Q(t)$ will completely recover to a situation where it is equal to, higher or lower than the initial state before attack. (a) shows the condition where the city network is fully recovered, with the network performance $Q(t^*)$ back to 100 %. (b) shows the condition where the city network is only partially recovered. Regional city network systems were impacted by the attack that could not be fully recovered and overall performance levels were reduced. (c) shows the condition where the city network completes a super-recovery after the attack. The network performance $Q(t^*)$ exceeds its original level. For example, the state of infrastructure in Haiti were even improved over pre-disruption levels after the Haiti earthquake in 2010.

3.2.1. Network performance indicator

The city network performance indicator $Q(t)$ is quantitative measured by the normal node ration, which has been widely used to measure the performance level of the network under the risk propagation (Basole & Bellamy, 2014). The calculation of network performance $Q(t)$ at period t is shown in Eq. (5), and n represents the total number of nodes in the city network.

$$Q(t) = \frac{\sum_{i=0}^n x_i(t)}{n} \quad (5)$$

3.2.2. Network resilience indicator

As shown in Fig. 4., During the whole attack-recovery period, the resilience of the network not only represents the ability to maintain original performance, but also reflects its ability to recover quickly from the attack. Therefore, we adopt the representative feature subset of three resilience indicators to measure this ability of the city network: the robustness loss, the recovery time and the loss of quality. Resilience indicators under these three dimensions are listed and defined in Table.3.

1. Robustness loss: robustness reflects the ability of a network to absorb disaster attack and maintain the original performance, which is constrained by both risk propagation and intrinsic recovery (Tierney & Bruneau, 2009). $\Delta Q = 1 - Q_{min}$ represents the network's

robustness loss after the disaster attack, indicating the ability of the network to resist disasters. Larger ΔQ indicates the more robustness loss of the network and the weaker resilience of the network.

2. Recovery time: resilience in network resilience is associated with the ability of the network to recover to the steady state in a timely manner (Tierney & Bruneau, 2009). In this paper, t^* which indicates the moment of recovery stability before T is used to measure the recovery time of the network. The longer the recovery time, the lower the recoverability and the poorer the resilience of the network.
3. Loss of quality: loss of quality reflects the fluctuation of network performance during the period from the attack interruption to the complete recovery of the system. This index effectively combines the robustness and recovery time, and takes the speed of network recovery into account (Ponomarov & Holcomb, 2009; Tierney & Bruneau, 2009). The total loss in the whole period, network performance can be expressed as the following formula: $L = \int_0^T [1 - Q(t)] dt$, the larger the L indicates network resilience ability is poorer.

4. Study area and data sources

With the deepening of reform and opening up, China's urbanization process has continued to advance over the past 40 years (Wu & Sun, 2018). The increasing connectivity of China's city network has also exacerbated the spread of the risk of public health attacks. In the face of public health disaster attack, the Chinese government tends to rely on city-to-city interactions to carry out coordinated emergency response and promote regional recovery. This resilience analysis study based on the coupling of city network risk propagation and cooperative recovery can better meet the needs of resilience analysis in the region of mainland China. Therefore, we selected mainland China as the study area (as shown in Fig. 5).

4.1. Data sources

We collected basic data sets of prefecture-level cities in China (as shown in Table.4) to construct Chinese city network $G(V, E)$. According to *China Statistical Yearbook 2020*, there are a total of 337 prefecture-level cities in mainland China. The distribution of cities in the region of mainland China varies greatly, with dense cities in the east but sparse cities in the west, and the distribution of population and medical resources in each city is also extremely uneven. These will lead to significant differences in the overall regional impact with different cities suffering the disaster attack.

4.2. Characteristics of double-layer coupling network

The action of the risk propagation and cooperative recovery mechanism in Chinese city network is shown in Fig. 6 (the example of Wuhan suffering a public health disaster attack). As shown in Table.5, in the risk

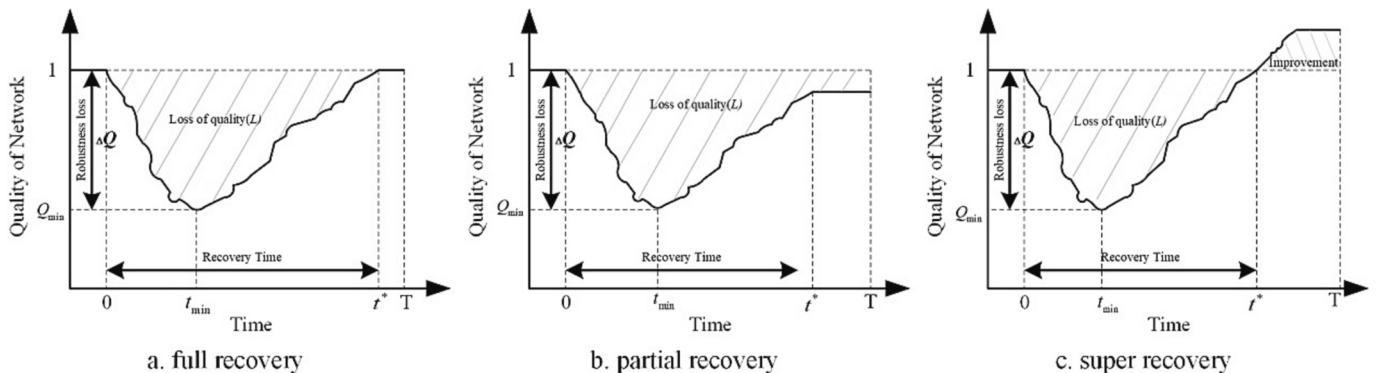


Fig. 4. Network resilience curve.

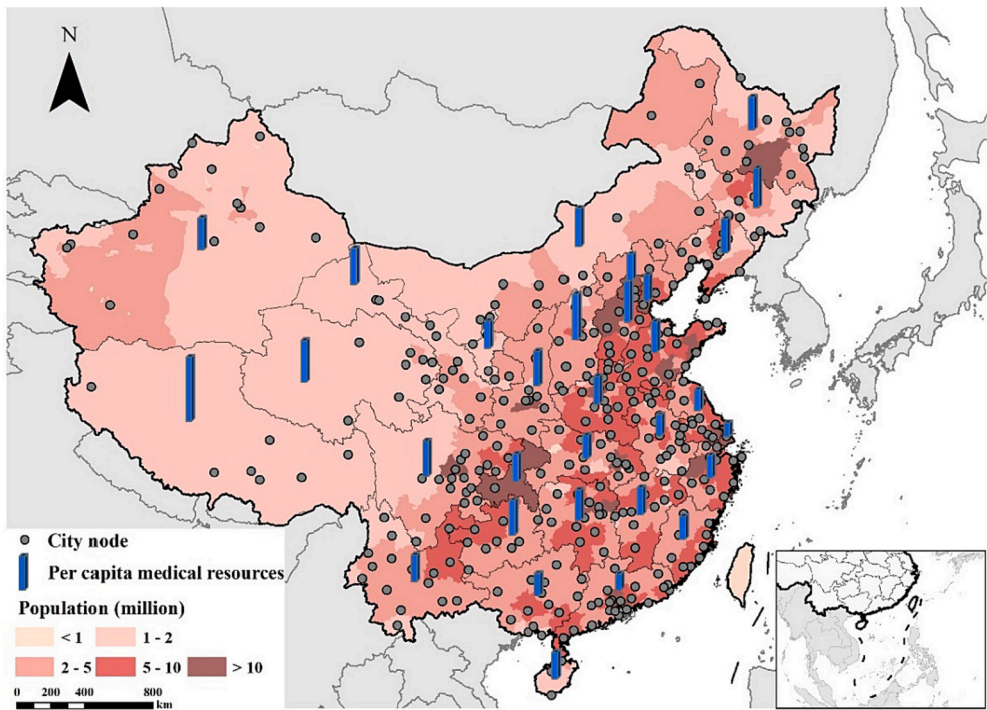


Fig. 5. Study area.

Table 4
Data sources.

Data type	Data sources
City node v_i	2020 Administrative division code of China (www.mca.gov.cn)(www.mca.gov.cn/article/sj/xzqh/1980/202105/20210500033655.shtml)
Population R_i	The seventh census of the National Bureau of Statistics of China(www.stats.gov.cn)
Medical resources S_i	China National Health Commission(www.nhc.gov.cn/)
Space distance of cities d_{ij}	AMAP(ditu.amap.com)
GDP of each city	China Statistical Yearbook 2020(www.stats.gov.cn)

propagation network, there are 853 links e_{ij}^a of risk propagation effect. The degree value of nodes in the network ranges from 1 to 10, and the average degree value is 5.07. The network is approximately a scale-free network. While in the cooperative recovery network, the damaged node will get emergency support from 336 other cities in the network, with a total of 112,896 edges e_{ij}^b , and the whole network is a fully connected network.

According to Eqs. (1)–(4), we calculated the potential risk propagation impact $U_d(i)$ and cooperative recovery impact $U_r(i)$ of each city in turn and then assign values to the damage probability $P_d(i)$ and recovery probability $P_r(i)$ of each city with $\tau = 50$. As shown in Fig. 7(a) and 7(b), in the Chinese city network, cities in the eastern region have higher risk propagation probability than cities in the western region, and cities with high damage probability show a trend of intra-regional clustering; cities in the central and western regions have better recovery probability than cities in the eastern region. Unlike cities with high damage probability clustered in core urban agglomerations, cities with higher recovery probability are more remote and have smaller population.

5. Resilience analysis

5.1. Experimental design

We simulated the change of network resilience of different cities suffering a sudden public health disaster attack to assess the resilience of the study region. During the experiment, we gave the following hypothesis descriptions for the disaster attack scenario:

- There is a single attack event—we assume that there is only a single impact on the network at the initial moment, and no other types of attacks are introduced in the process of risk propagation and cooperative recovery.
- The disaster attack happens at the node-level—we only consider the state changes after nodes suffer attacks. Disaster events only affect the individual city, ignoring the impact on the correlation between cities.
- The state of each node is known with certainty—for the purposes of characterizing the effects of risk propagation and cooperative recovery, we assume that the current state of each node is known at any given point in period.
- The region city network performance $Q(t)$ will reach full recovery—Within a certain period after the strike, all cities will eventually return to the normal state after taking corresponding preventive and control measures. To better quantify the resilience, we assume that all cities will reach the normal state within the recovery time, with the region city network performance $Q(t) = 1$, and the region has fully recovered.

During the experiments, we selected one city in the network in turn as the node of the city where a sudden public health disaster event broke out, and used 200 periods to simulate the fluctuation of the whole network performance after the attack. We conducted 100 attack simulations for each city, collecting a total of 33,700 experimental data, and take the average value of the network performance to plot the corresponding resilience curve to average the random effect.

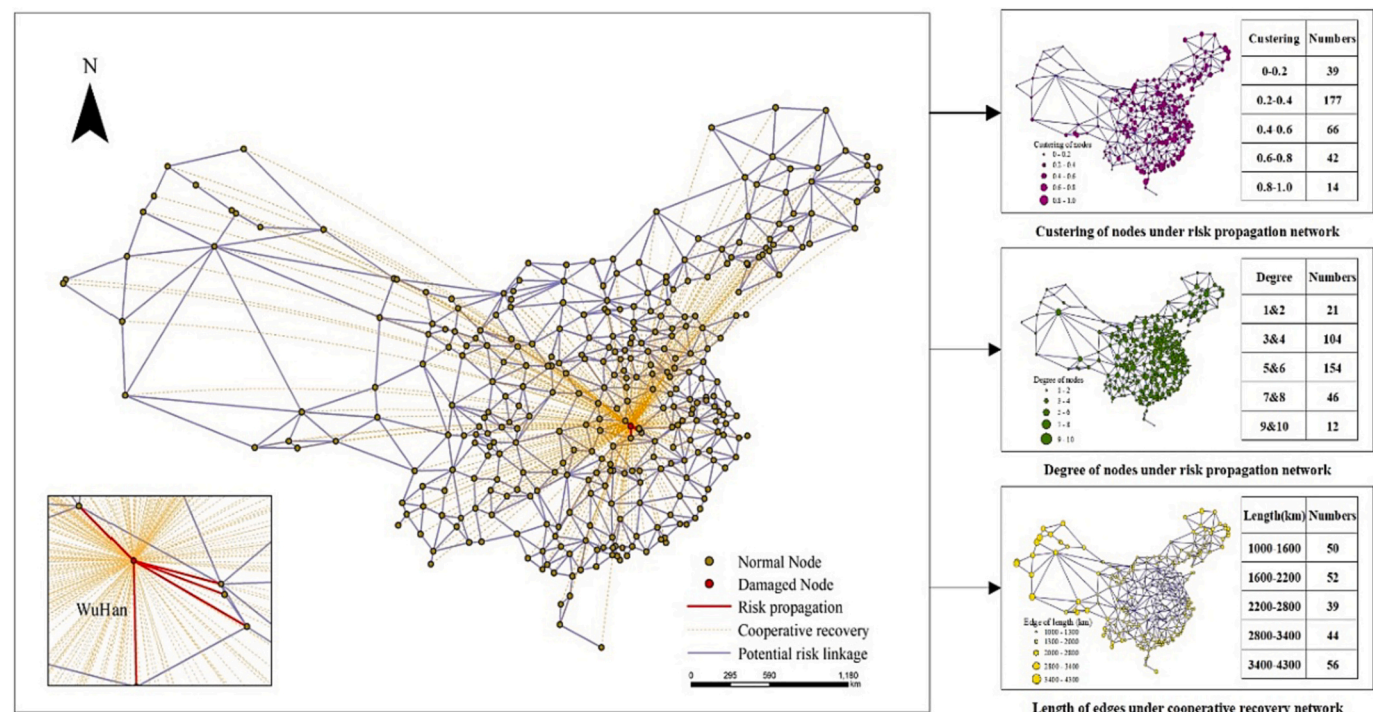


Fig. 6. Chinese city network (disaster attack to Wuhan).

Table 5
Basic data of city network topology.

Network topology feature	Risk propagation	Cooperative recovery
Number of nodes	337	337
Number of edges	853	112,896
Average degree	5.07	336
Average clustering	0.41	1
Average edge length(km)	452	1541

5.2. Clustering of nodes based on the impact on network resilience

In each simulated attack, the overall network performance $Q(t)$ shows a degradation-recovery process. Fig. 8(a) shows the thermal distribution of the network performance state at various time periods after

disaster attack. During the 0–20 and 120–200 time periods, most of the network performance under the attack is in a better stable state, while during the 20–40 periods, most of the network performance will decrease to below 0.86, which means the disaster attack will have a greater impact on the network resilience at these periods. Fig. 8(b) shows the maximum and minimum network performance fluctuating condition of the impact of disaster attacks on the three dimensions of network resilience. When different cities suffer disaster attacks, they have different impacts on the whole regional resilience, with the robustness loss, the recovery time and the loss of quality all showing difference. Therefore, the regional resilience can be effectively promoted through the reinforcement of disaster-resistant construction in cities that have significant impact in disasters.

In further study, all cities are analyzed by clustering them according to their different impacts on the three dimensions of network resilience

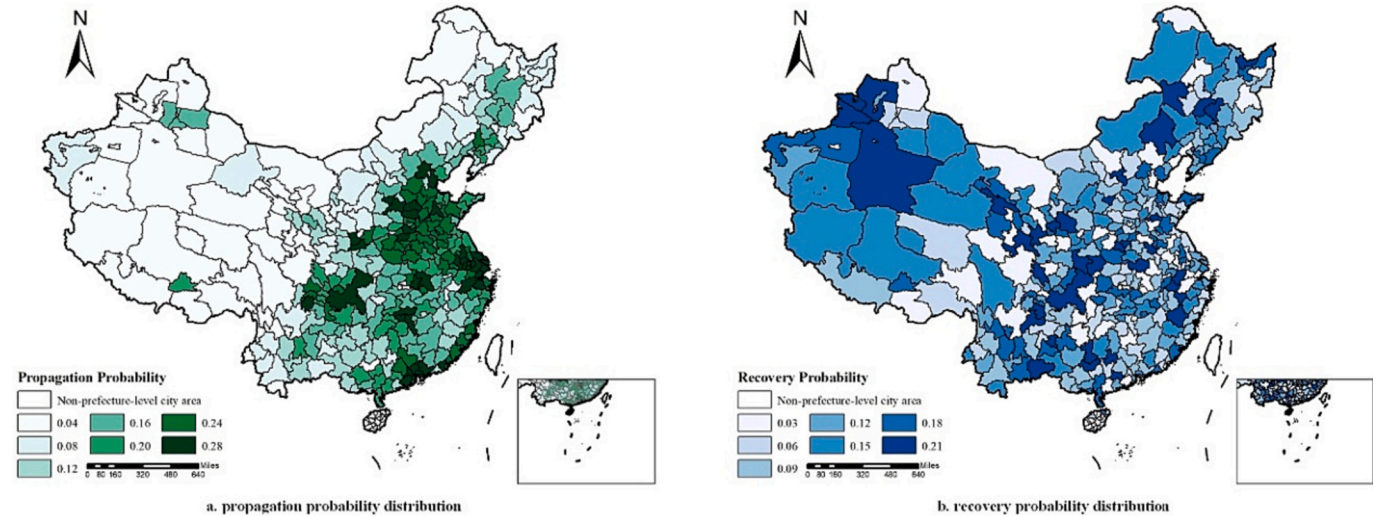


Fig. 7. Urban distribution of $P_d(i)$ and $P_r(i)$

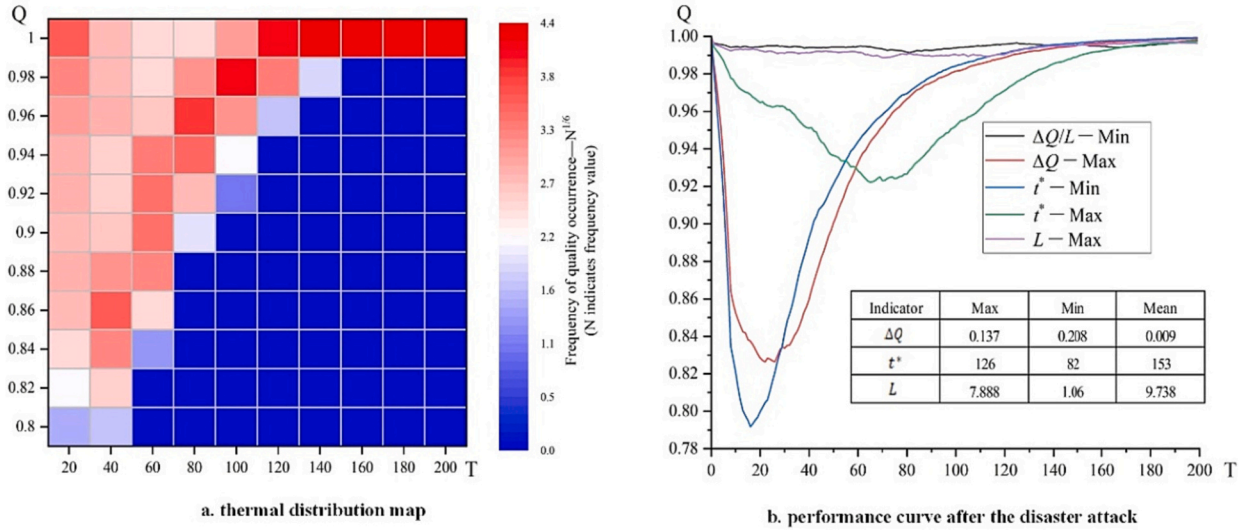


Fig. 8. Network performance fluctuates under disaster.

after a sudden public health disaster attack. We standardized the three indicators of network resilience and analyzed them by K-means clustering method (Haciefendioglu et al., 2022; Kanungo et al., 2002). All the cities are divided into 5 categories.

The results are shown in Fig. 9 and Fig. 10. In the Chinese city network, the forth category of cities (e.g., Shenyang, Shenzhen, etc.) and the fifth category of cities (e.g., Shanghai, Tianjin, etc.) are the most numbers and distributed centrally in the subnetwork. After these cities suffer the disaster attack, the overall network resilience capacity is significantly affected in the three dimensions, and the resilience capability becomes lower, resulting in serious losses. On the other hand, there are only 5 cities belonging to the first category (e.g., Aksu, etc.). The city network still has the best resilience in the three dimensions after these cities suffer the disaster attack. In terms of the spatial distribution analysis of the cities in each category, the western and northern cities generally belong to the top three categories, which have smaller impact on the resilience of the network under the disaster attack. While cities in the southeastern region generally majority fall into the last two categories, which have a greater impact on the resilience of the overall network after disasters.

The results means that, in the face of sudden public health disaster attacks, the central and eastern regions of China with denser cities and more concentrated populations was hit with a more severe impact on the

overall resilience of the region. The west and northeast region with sparser cities and smaller populations is less resilient to the whole city network. In reality, two most impactful public health disaster attack in China's history both erupted in the fifth category of cities in the central or eastern regions. The COVID-19 epidemic that broke out in Wuhan in early 2020 and the SARS epidemic that broke out in Guangzhou in 2003 both spread rapidly and wreaked havoc across the entire region of China (Yang et al., 2020). According to Wu et al.'s study of spatio-temporal evolution of COVID-19 epidemic in China (Wu et al., 2021), the hot spot of epidemic area mainly distributed in the central and eastern regions. And the cold spot of epidemic area mainly distributed in the southwest northwest and northeast regions. This is consistent with the spatial distribution of city resilience impacts in the results of the sudden public health disaster attack simulation experiment.

Therefore, it is necessary to strengthen the capability of the corresponding cities in the central and eastern regions of China to protect themselves from sudden public health disaster attacks, reducing the likelihood of public health emergencies at their source. When a public health emergency occurs in these category 5 cities in the central and eastern region, an emergency response should be carried out immediately to weaken its population flow links with surrounding cities to reduce the spread of risk, while at the same time taking full advantage of the strong cooperative capacity between these cities to reduce recovery

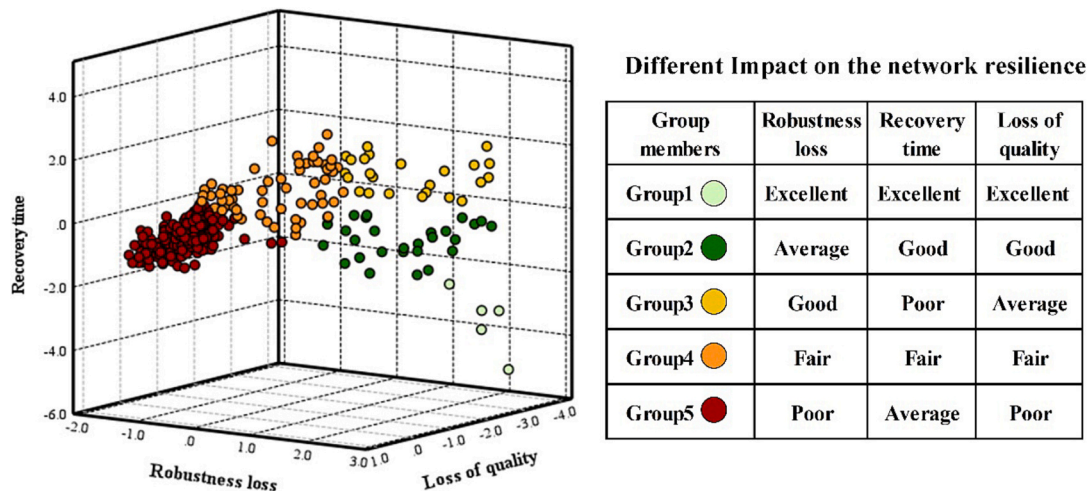


Fig. 9. Node cluster analysis diagram.

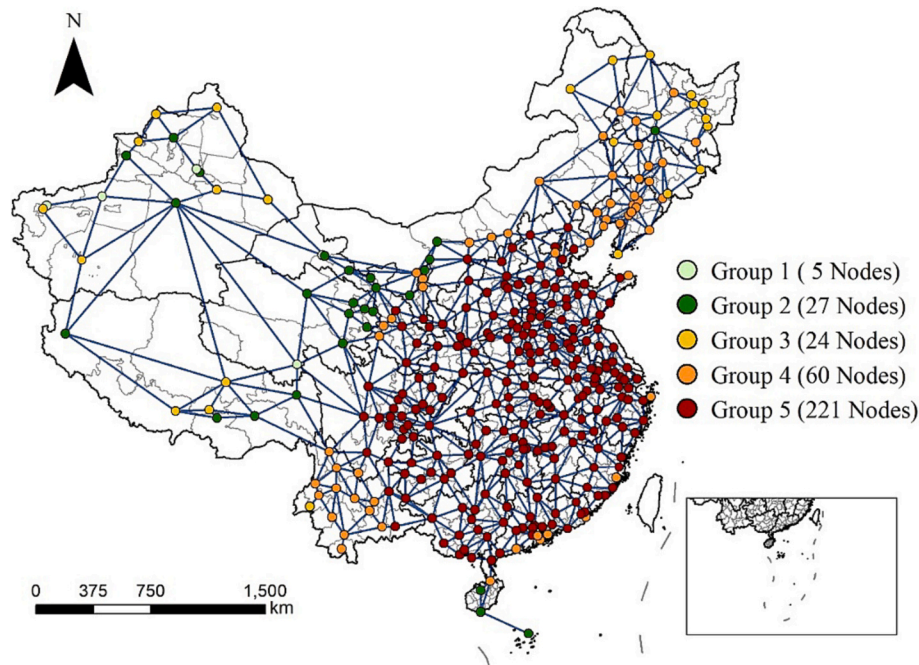


Fig. 10. Node influence clustering distribution diagram

time. While for the prevention of sudden public health attacks southwest northwest and northeast regions, the rapid realization of the synergistic mutual assistance of medical resources and the enhancement of transportation to realize the cooperative recovery action based on the city network should be the focus of attention.

5.3. Correlation between influence factors and resilience

In this section, six possible influencing factors of regional city network are selected (as independent variables) to explore the correlation of different factors and different indicators of network resilience (as dependent variables).

5.3.1. Independent variables

The independent variables in the city network resilience analysis are associated with network topological feature and urban development level. Degree, clustering and edge length are important indicators of network topological feature. Degree reflects the importance of nodes to the network. Clustering describes the degree of closeness between other nodes adjacent to a node and edge length reflects the weight of the connected edge where the node is located in the network. On the other way, population, GDP and medical resources of a city are important indicators of urban development level. Population shows the scale of a city. GDP reflects the economic level and medical resources reflect the medical level of the city to cope with the attack. Therefore, degree (D), clustering(C), edge length(E), GDP(G), population(P) and medical resources (MR) are selected as independent variables in the correlation study of network resilience factors.

5.3.2. Dependent variables and models

The dependent variables in the resilience influence analysis are the three indicators of resilience capacity: Robustness loss— ΔQ , Recovery time— t^* , Loss of quality— L . To investigate the relationship between independent variables and these dependent variables, we use MLR (multiple linear regression) model, as it is able to evaluate relationship between a set of correlated dependent variables and different sets of exogenous explanatory variables. Below are the detailed model specifications:

$$\Delta Q = \beta_{10} + \beta_{11}D + \beta_{12}C + \beta_{13}E + \beta_{14}G + \beta_{15}P + \beta_{16}MR$$

$$t^* = \beta_{20} + \beta_{21}D + \beta_{22}C + \beta_{23}E + \beta_{24}G + \beta_{25}P + \beta_{26}MR$$

$$L = \beta_{30} + \beta_{31}D + \beta_{32}C + \beta_{33}E + \beta_{34}G + \beta_{35}P + \beta_{36}MR$$

5.3.3. Correction analysis

Table.6 shows the descriptive statistics of the variables. We run our analyses in IBM SPSS Statistics 27 and present MLR regression results for the different network resilience capacity in Table.7, from which we summarize Table.8 To illustrate the correlation between different factors and the influence after disaster event of network resilience.

Standard errors in parentheses * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

The results show that for robustness loss of the network resilience, degree in network topology and population in urban development level are positively correlated with network robustness loss, while edge length in network topology and GDP and medical resources in urban development level are negatively correlated with robustness loss. For the loss

Table 6
Descriptive statistics.

	N	Mean	Standard Deviation	Min	Max
Independent variables					
Degree (risk propagation)	337	5.07	1.74	1	10
Clustering (risk propagation)	337	0.41	0.20	0	1
Edge length(km) (Cooperative recovery)	337	1536.35	533.79	1051.60	4295.60
GDP (*10 ⁸)	337	2993.28	4576.53	0	38,701
Population(*10 ⁴)	337	415.24	376.30	0.23	3205.42
Medical resources (*10 ⁴)	337	454.26	405.91	0.24	3388.00
Dependent variables					
ΔQ	337	0.137	0.049	0.208	0.009
t^*	337	126	9	82	153
L	337	7.888	2.070	1.060	9.738

Table 7
Multiple regression analysis of resilience.

Indicator		ΔQ	t^*	L
Constant		0.205*** (<0.001)	114.467*** (<0.001)	9.801*** (<0.001)
Independent Variables				
Network topology	Degree	0.004*** (<0.001)	-0.323 (0.325)	0.209*** (<0.001)
	Clustering	0.01 (0.216)	8.378** (0.001)	1.286*** (<0.001)
	Edge length	-0.000*** (<0.001)	0.006*** (<0.001)	-0.003*** (<0.001)
	GDP	-0.000*** (<0.001)	-0.000 (0.966)	-0.000** (0.006)
Urban development level	Population	0.000*** (<0.001)	-0.001 (0.821)	0.003*** (<0.001)
	Medical resources	-0.000*** (<0.001)	0.003 (0.401)	-0.001* (0.011)
Number of observations		337	337	337
R-squared		0.721	0.141	0.720
p-value		<0.001	<0.001	<0.001

Table 8
Direct impact of independent variables.2.

Factors		ΔQ	t^*	L
Network topology	Degree	+(***)	No	+(***)
	Clustering	No	+(**)	+(***)
	Edge length	-(***)	+(***)	-(***)
	GDP	-(***)	No	No
Urban development level	Population	+(***)	No	+(***)
	Medical resources	-(***)	No	-(***)

of quality in network resilience, degree, clustering in network topology and population in urban development level are positively correlated with it, while edge length in network topology and GDP and medical resources in urban development level are positively correlated with it. However, the recovery time in network resilience is only affected by clustering and edge length in network topology and is positively correlated with it.

According to the above analysis, the corrections between the impact of each factor and the three dimensions of network resilience indicators are different. Each influencing factor may only correlate with one or two dimensions of network resilience indicators. In addition, according to the correlation between edge length and each dimension of network resilience, there is a trade-off impact between robustness loss and recovery time for the whole regional resilience. When the robustness loss of the total network is maintained well, it may need to take relatively longer time to complete the recovery.

Therefore, when preventing the impact of sudden public health disasters attack on the city network, it is necessary to effectively balance the three dimensions of resilience capability to achieve the optimal condition. For sudden public health disasters with a high capacity to spread infection, particular concern should be added to impact factors that are positively correlated with robust losses, such as cities with higher degree values and larger populations, with an emphasis on strengthening the defensive capacity of such cities. For public health events of longer duration, it is more important to focus on improving resilience in the dimension of recovery, with primary consideration given to cities with higher impact factors positively correlated with recovery time.

6. Conclusion

In this paper, we propose a methodology framework for assessing the

resilience of regional city networks affected by risk propagation and collaborative recovery. In terms of theoretical model, we view the region as a two-layer coupled city network and characterize its resilience capacity after suffering disaster attack through three dimensions: robustness loss, recovery time and the loss of quality, which enriches the analytical model for overall network resilience. By constructing China's prefecture-level city network, we conducted simulation experiments on the network resilience of different cities when they were attacked by a sudden public health disaster. Through the clustering analysis of resilience assessment indicators, we found that there are significant differences in the resilience performance of cities in different regions of China, which implies that emergency response capacity construction should be systematically designed by region. In addition, six factors including degree, clustering, edge length, GDP, population and medical resources were selected to study the potential influencing factors of the city network resilience, and the correlation between the three indicators of resilience and various factors was obtained. For the city nodes with high degree, clustering in the network which are in the core area and city nodes in the network which have a higher population, there is a need to further strengthen disaster preparedness for them to reduce impact on the overall regional resilience after suffering disaster. In addition, further increasing the city's medical resources and GDP levels could also help reduce the impact on the overall regional resilience.

However, the simulation of network attack in this study is still limited to the attack of node states. In order to enrich the fluctuations of network performance in the framework of network resilience analysis, the impact on other components such as network edges under the disaster attack requires further in-depth study. Meanwhile, our study found that in Chinese city networks, the network topology and the regional development level have a significant impact on the overall network resilience under the disaster attack. So, focusing on different urban clusters in the city network will be another major research direction to understand regional resilience. In addition, for improving the resilience of the region to cope with the disaster strikes of public emergencies, the level of government governance, including the allocation, scheduling, and management of medical resources, public opinion control, and comprehensive community governance, etc., will likewise be important influencing factors. The differences in the level of government governance will be an important new direction in the comparison of the resilience of different small regions.

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CRedit authorship contribution statement

Kou Longbin: Conceptualization, Data curation, Formal analysis, Funding acquisition, Resources, Writing – original draft, Writing – review & editing. **Zhao Hanping:** Funding acquisition, Investigation, Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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